Week4

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1 Sarsa Learning Update:

It is an "on-policy" algorithm as it updates the Q-value based on the action a_{t+1} which is chosen by the current policy (e.g.$\epsilon - greedy$ policy).

The update rule is:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(\Delta)$$

Where Δ is the T.D. error.

$$\Delta = TD\ Target - Current\ Value$$

$$TD\ Target = \mathcal{R} + \gamma Q(s_{t+1}, a_{t+1})$$

Where a_{t+1} is chosen using the current policy from state s_{t+1}.

Therefore, the update is:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(\mathcal{R} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t))$$

This is done for each non terminal state for all the episodes. The Q-values converge to the optimal Q-values given:

- All state-action pairs must be visited an infinite number of times. This is typically ensured by using an exploration strategy like ϵ-greedy.
- The policy must converge in the limit to a greedy policy. This means that the exploration parameter (ϵ) must be gradually reduced toward zero over time.

2 Q-Learning Update:

It is an "off-policy" algorithm since the behaviour policy (used to gerenate behaviour) is different from estimation policy (the one being evaluated and improved).

The behaviour policy should be exploratory (like $\epsilon - greedy$) and the estimation policy is greedy.

So the update is:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(\Delta)$$

$$\Delta = \mathcal{R} + \gamma max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)$$

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(\mathcal{R} + \gamma max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t))$$

3 Conclusion:

Use Q-Learning if you want to find the theoretical optimal strategy and can afford to explore safely; use SARSA if your agent needs to learn a safe strategy that accounts for the reality of its exploration noise.